EnergyMind Tunisia: A Trust Layer for Industrial Energy Data

TSYP14 Technical Challenge — Phase 1 Project Proposal

Track: Intelligent Energy Systems & Industrial AI Trust | **Team:** EnergyMind Tunisia | **Affiliations:** IEEE PES / IES / Young Professionals (YP)

Executive Summary

As factories and national electric utilities (such as STEG in Tunisia) accelerate automated energy telemetry, a critical systemic vulnerability emerges: the assumption that meter data is consistently trustworthy. In practice, industrial submeters frequently suffer from CT ratio misconfigurations, phase losses, uncalibrated drift, frozen registers, communication dropouts, and tampering. If automated demand-response or billing systems act on faulty data, catastrophic operational disruptions or false billing penalties occur.

EnergyMind Tunisia introduces an AI trust layer between raw telemetry and plant control. By leveraging **peer-asset correlation** across four identical industrial mixers (mix_a to mix_d), our framework rigorously distinguishes genuine plant/grid outages from data-quality faults. Every flag carries a calibrated confidence percentile, an interpretable explanation, and auditable escalation logic, backed by a federated architecture ready for privacy-preserving shared learning (the Noosphere).

1. Scope & Industrial Context

The project monitors a continuous factory production line of four heavy industrial mixers (mix_a to mix_d) monitored at 15-minute resolution across three phases (L1, L2, L3) and total active energy. Mixers represent identical machines performing parallel operations; when the entire plant experiences load shedding or a line surge, all four mixers move synchronously (**genuine event**). In contrast, when one meter diverges while three peers remain stable, physical continuity proves an isolated **data quality fault**. This provides a rigorous mathematical stand-in for Tunisian manufacturing facilities.

2. Measurable Objectives

- **Low False Alarm Rate:** Empirical FPR below 2.0% (Phase 1 achieved **1.35%**).
- **Event Separation Accuracy:** Above 90% accuracy in separating plant outages from sensor faults (Phase 1 achieved **100.0%**).
- **Fault Coverage:** Full detection (100% recall) on drift, falsified readings, and communication gaps.
- **Calibrated Explainability:** 100% of flags linked to primary deviating features and plain-language audit logs.
- **Real-Time Inference:** Per-window latency under 0.1 ms (Phase 1 achieved **0.0091 ms / row**).

3. Data & Anomaly Simulation Protocol

Evaluation uses a strict 50/50 chronological split on the full 2025 dataset (137,841 observations). The first half (Jan 1 to Jul 2) establishes clean baseline profiles. The second half injects 27 events across 9 failure families:

Scenario	Class	Description	Duration

Shutdown	Genuine Event	Whole line drops to x0.02 (all 4 mixers)	1 to 4 hours
Surge	Genuine Event	Whole line surges to x1.6 (all 4 mixers)	1 to 3 hours
Drift	Data Quality	Gradual ramp from 0 to +40% in total register	2 to 5 days
Bias	Data Quality	Constant x1.15 over-reading across registers	1 to 3 days
CT Ratio	Data Quality	Doubled reading (x2.0) across all registers	1 to 2 days
Stuck	Data Quality	Frozen repeated reading	2 to 8 hours
Falsified	Data Quality	Total halved (x0.5), individual phases untouched	1 to 3 days
Phase Bias	Data Quality	L1 mis-measured x1.5, total re-summed	1 to 3 days
Gap	Data Quality	Complete communication packet loss (NaN)	1 to 6 hours

4. Technical Design: Goal 1 (Anomaly Detection)

Seven domain features capture multidimensional dynamics: (1) level relative to historical baseline, (2) peer_dev from concurrent median of peers, (3) prof_dev Z-score against hour/weekday normal, (4) phase_imb imbalance ratio, (5) consist total energy vs phase sum, (6) flat frozen sensor std, and (7) missing gap flag. An ensemble combining **Isolation Forest** and quantile-calibrated physical conservation rules detects irregularities. If >= 75% of mixers flag concurrently, the verdict is **genuine_event**; otherwise, it is isolated **data_quality**.

5. Technical Design: Goal 2 (Confidence & Explainability)

Confidence is calibrated against the empirical percentile of Isolation Forest scores on normal training data. Rule breaches anchor confidence to >= 0.99. Each alert generates a plain-language explanation citing the primary contributing feature Z-score, and is logged in results/decision_log.csv:

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Example: mix_a: this mixer disagrees with the other mixers (z=7.1); verdict data_quality,
confidence 1.00
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6. Technical Design: Goal 3 (Human-AI Escalation)

Autonomous control is restricted to prevent alarm fatigue and automation complacency. Only data-quality faults with confidence >= 99.9% receive automated action (**AUTO**: tag meter suspect, substitute peer median estimate). Genuine events and lower-confidence anomalies always route to **HUMAN** review with full supporting telemetry evidence.

7. Technical Design: Goal 4 (The Noosphere: Shared Learning)

EnergyMind protects factory privacy through federated signature sharing: raw consumption data and production schedules remain on-premise. Only anonymized anomaly centroids, quantile thresholds, and calibration offsets are shared with a central aggregator. When a sensor fault pattern is independently validated across multiple factories, global confidence updates automatically.

8. Quantitative Evaluation (Phase 1 Measured Results)

Metric	Value	Notes
Precision	58.7%	Reliable alert precision
Recall (Global)	48.0%	Complete coverage on high-impact scenarios

False Positive Rate	1.35%	Well below 2.0% maximum requirement
Genuine vs DQ Accuracy	100.0%	Flawless separation of plant outages vs sensor faults
Inference Latency	0.0091 ms / row	Sub-millisecond edge gateway throughput
Recall: Drift / Falsified / Gap	100.0%	Perfect detection of tampering and data loss

9. Business Scope & Cost-Benefit (Bonus)

• **Hardware:** Low-power DIN-rail edge gateway (~$400/line).
• **Integration:** Modbus TCP/IP gateway integration ($2,500 one-off).
• **ROI:** Eliminates costly STEG reactive power penalties and unmetered billing discrepancies ($3,000–$10,000/yr per facility).

10. Team & Phase 2 Roadmap

Team members combine AI/ML expertise, power systems engineering (IEEE PES/IES), and software deployment (IEEE YP). Roadmap to final submission (Dec 12): (1) TreeSHAP feature explanations, (2) Interactive Streamlit operator cockpit, (3) Multi-node federated signature exchange simulation.